ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)
ORGANISATION OF ISLAMIC COOPERATION (OIC)
DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING

Practice Examination (Self-Study)

Course No.: Phy 4143 | Time: 2 Hours

Course Title: Physics II | Full Marks: 100

Scope: Lecture 1–3 only (Basic Concepts, Basic Laws, Y-Δ Transformation)

There are 3 (three) questions. Answer all questions. The symbols have their usual meanings. Marks of each part are written in brackets. Draw clear circuit sketches where helpful for your own working (this practice paper describes circuits in words since it was generated for self-study, not handed out in class).

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1. Theory and Definitions

a) Define voltage and current. State Ohm's Law with its formula. [3]

b) Define an electric circuit. State Kirchhoff's Current Law (KCL) and Kirchhoff's Voltage Law (KVL). [4]

c) Define node, junction, branch, loop and mesh. Clearly explain the difference between a node and a junction. [5]

d) Differentiate between an ideal independent voltage source and an ideal independent current source, including their internal resistance. [4]

e) State the condition under which source transformation is valid, and write the transformation formula in both directions (voltage source to current source, and vice versa). [4]

f) State the rule for combining resistors in series and in parallel. Write the formula for R_eq of n series resistors and of two parallel resistors. [4]

g) State the Voltage Divider Rule and the Current Divider Rule with their formulas, clearly defining every symbol used. [4]

h) Briefly explain, with an example situation, when a Y-Δ (or Δ-Y) transformation becomes necessary in resistive network analysis. [2]

2. Basic Laws — Ohm's Law, KVL, KCL, Source Transformation

a) A 30 V source is connected in series with three resistors R1 = 2 Ω, R2 = 3 Ω and R3 = 5 Ω (a single loop). Find the loop current and the voltage across each resistor. Verify your result using KVL. [8]

b) Two voltage sources, 18 V and 6 V, both oriented to aid each other, drive a single-loop series circuit containing a 3 Ω and a 5 Ω resistor. Find the loop current and the voltage across each resistor. [8]

c) At a single node a, four branches are connected in parallel, each returning to ground: an 8 A independent current source entering node a; a 4 Ω resistor carrying current io (defined flowing from node a to ground); a dependent current source of value io/2 (same reference direction as io); and an 8 Ω resistor with voltage vo defined across it (+ at node a). Use KCL at node a to find io and vo. [10]

d) A practical voltage source consists of a 20 V ideal source in series with a 5 Ω resistor. (i) Find its equivalent current-source representation. (ii) This equivalent current source (with its 5 Ω in parallel) now feeds a 15 Ω load resistor. Use the current divider rule to find the current through the load, and the voltage across it. [9]

3. Resistor Connections, Dividers and Y-Δ Transformation

a) Between terminals A and B: a 4 Ω resistor is in series with the parallel combination of a 6 Ω and a 3 Ω resistor, and this whole combination is in series with a 5 Ω resistor. Find R_eq between A and B. [6]

b) Three resistors R1 = 4 Ω, R2 = 6 Ω, R3 = 10 Ω are connected in series across a 40 V source. Using the Voltage Divider Rule, find the voltage across R2. [6]

c) Three resistors R1 = 6 Ω, R2 = 12 Ω, R3 = 4 Ω are connected in parallel, and a total current of 18 A enters the parallel combination. Using the Current Divider Rule, find the current through R1. [7]

d) A Y-connected (star) resistor network has R1 = 4 Ω, R2 = 6 Ω, R3 = 12 Ω (R1, R2, R3 connect the star point to terminals a, b, c respectively). Convert this network to its equivalent Δ (delta) network, i.e. find R_A, R_B and R_C. [8]

e) Using the Δ network you obtained in part (d), convert it back to a Y network and verify that you recover the original values of R1, R2 and R3. [8]

Answer Key

(Final answers only — try every question first before checking.)

Q2.

a) i = 3 A, v1 = 6 V, v2 = 9 V, v3 = 15 V (sum = 30 V, checks out)

b) i = 3 A, v(3Ω) = 9 V, v(5Ω) = 15 V

c) io = 4 A, vo = 16 V

d) (i) Is = 4 A with 5 Ω in parallel (ii) i(load) = 1 A, v(load) = 15 V

Q3.

a) R_eq = 11 Ω (6Ω parallel 3Ω = 2Ω, then 4+2+5)

b) v2 = 12 V

c) i1 = 6 A (R_parallel = 2 Ω)

d) R_A = 36 Ω, R_B = 24 Ω, R_C = 12 Ω

e) R1 = 4 Ω, R2 = 6 Ω, R3 = 12 Ω — matches the original Y values, checks out